

KENDRIYA VIDYALAYA SANGATHAN, RAIPUR REGION

FIRST PRE-BOARD EXAM (2024-25)

CLASS-XII

SUBJECT – PHYSICS (042)

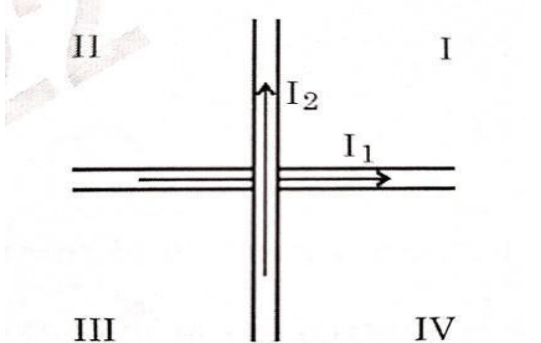
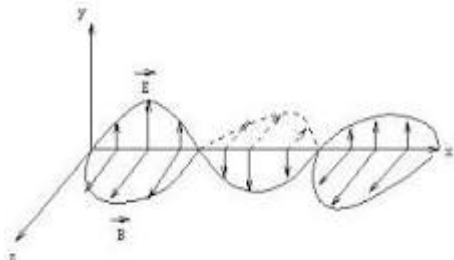
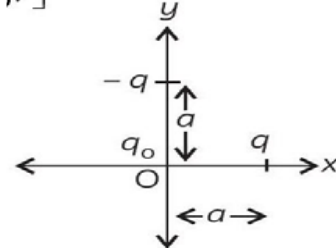
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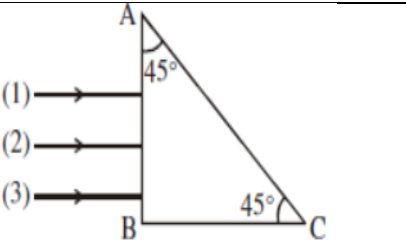
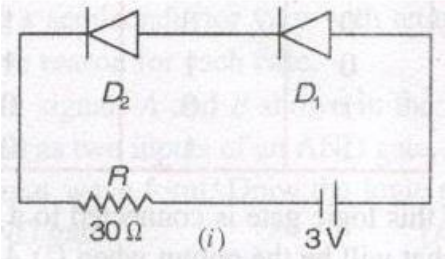
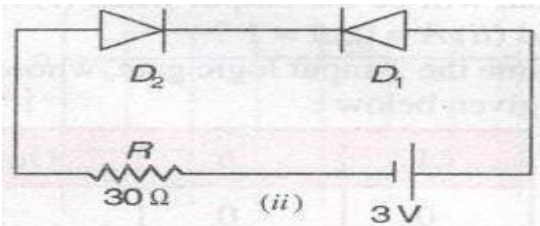
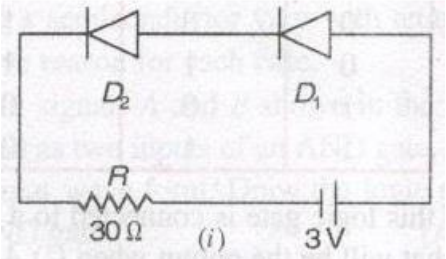
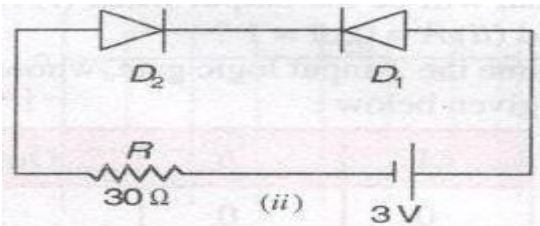
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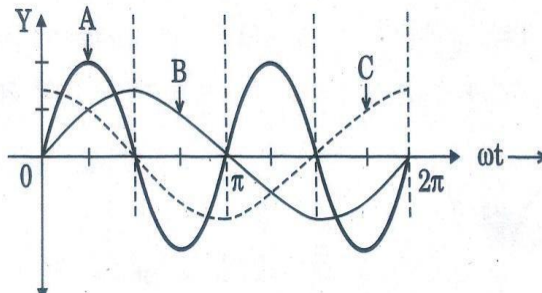
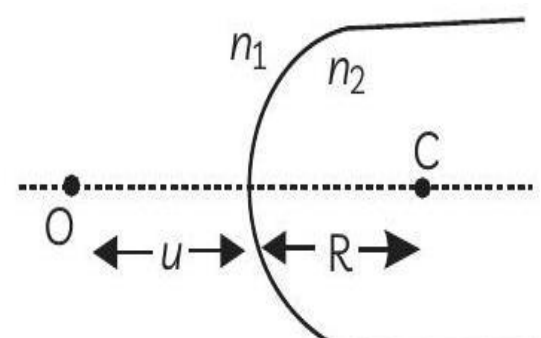
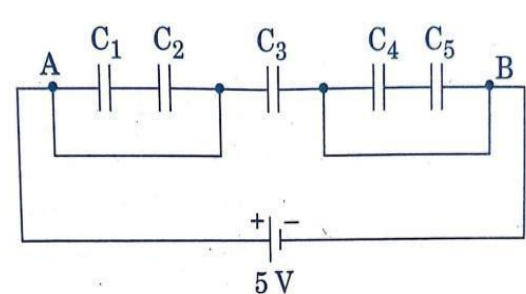
GENERAL INSTRUCTIONS:-

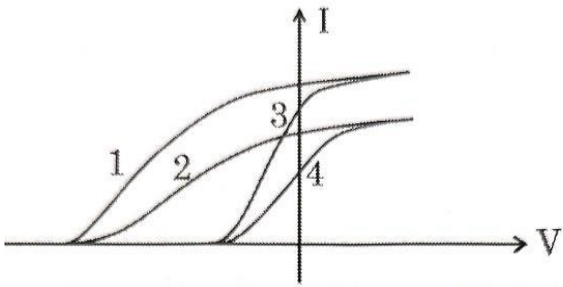
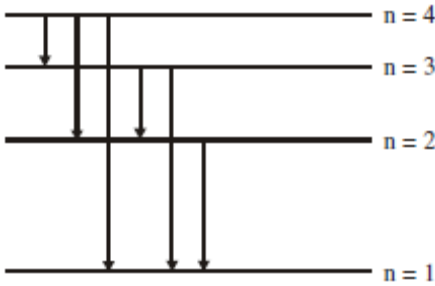
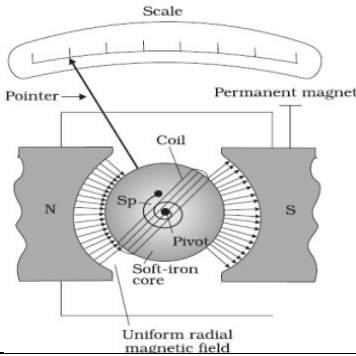
- (1) There are 33 questions in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- (3) All the sections are compulsory.
- (4) **Section A** contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, **Section B** contains five questions of two marks each, **Section C** contains seven questions of three marks each, **Section D** contains two case study based questions of four marks each and **Section E** contains three long answer questions of five marks each.
- (5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- (6) Use of calculators is not allowed.
- (7) You may use the following values of physical constants where ever necessary
 - i. $c = 3 \times 10^8$ m/s
 - ii. $m_e = 9.1 \times 10^{-31}$ kg
 - iii. $e = 1.6 \times 10^{-19}$ C
 - iv. $\mu_0 = 4\pi \times 10^{-7}$ TmA⁻¹
 - v. $h = 6.63 \times 10^{-34}$ Js
 - vi. $\epsilon_0 = 8.854 \times 10^{-12}$ C²N⁻¹m⁻²
 - vii. Avogadro's number = 6.023×10^{23} per gram mole

	SECTION-A	(16x1=16 marks)	
Q.1	In a circuit the phase difference between the alternating current and the source voltage is $\pi/2$. Which of the following cannot be the element (s) of the circuit? (a) only C (b) only L (c) L and R (d) L or C		1
Q.2.	The current in a device varies with time t as $I = 6t$, where I is in mA and t is in s. The amount of charge that passes through the device during t = 0 s to t = 3 s is (a) 10 mC (b) 18 mC (c) 27 mC (d) 54 mC		1
Q.3.	The susceptibility of a magnetic material is -4.2×10^{-6} . The material is: (a) ferromagnetic (b) paramagnetic (c) diamagnetic (d) none of the these		1

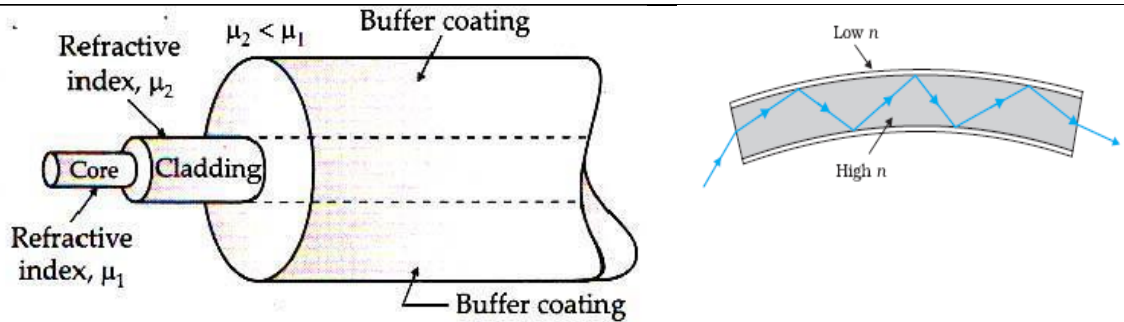
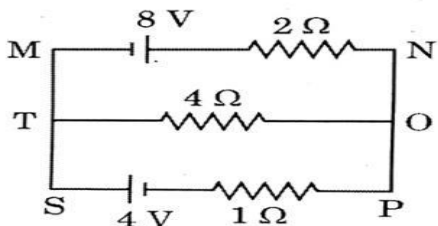
Q.4.	Two wires carrying currents I_1 and I_2 lie, one slightly above the other, in a horizontal plane as shown in figure. The region of vertically upward strongest magnetic field is (a) I (b) II (c) III (d) IV		1
Q.5.	A welder wears special glasses to protect his eyes mostly from the harmful effect of: (a) very intense visible light (b) infrared radiation (c) ultraviolet rays (d) microwaves		1
Q.6.	A variable capacitor is connected to a 200 V battery. If its capacitance is changed from $2 \mu\text{F}$ to $X \mu\text{F}$, then decrease in energy of the capacitor is $2 \times 10^{-2} \text{ J}$. The value of X is – (a) $1 \mu\text{F}$ (b) $2 \mu\text{F}$ (c) $3 \mu\text{F}$ (d) $4 \mu\text{F}$		1
Q.7.	Which of the following is/are true for electromagnetic waves?  I. They transport energy. II. They have momentum. III. They travel at different speeds in air depending on their frequency. (a) I and III (b) II only (c) I, II and III (d) I and II		1
Q.8.	Consider the fusion reaction: ${}^4\text{He} + {}^4\text{He} \rightarrow {}^8\text{Be} + Q$ For the reaction, Atomic mass of ${}^8\text{Be}$ is 8.0053 u and that of ${}^4\text{He}$ is 4.0026 u. & $1\text{u} = 931 \text{ MeV}/c^2$ then Q-value will be (a) -9.31MeV (b) -93.1MeV (c) -9.31keV (d) -93.1keV		1
Q.9.	Three charges q , $-q$ and q_0 are placed as shown in figure. The magnitude of the net force on the charge q_0 at point O is $k = 1/4\pi\epsilon_0$ (a) 0 (b) $2kqq_0/a^2$ (c) $\sqrt{2kqq_0/a^2}$ (d) $kqq_0/\sqrt{2a^2}$		1
Q.10	Two nuclei have their mass numbers in the ratio of 1 : 27. What is the ratio of their nuclear densities? (a) 1 : 27 (b) 1: 1 (c) 3: 1 (d) 1: 3		1

Q.11	Three rays (1,2,3) of different colours fall normally on one of the sides of an isosceles right angled prism as shown. The refractive index of prism for these rays is 1.39, 1.47 and 1.52 respectively. In which of these rays get internally reflected from surface AC (a) 1 & 2 (b) 1 & 3 (c) 2 & 3 (d) only 1		1
Q.12.	When similar ideal diodes D1 and D2 are connected as shown figure. The currents through the resistance R of the circuits (i) and (ii), will be   (a) 10A & 0.1 A (b) 0.1 A & zero (c) 0.1 A , 0.1A (d) 10 A , 10 A	 	1
	For Questions 13 to 16, two statements are given –one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below. a) If both Assertion and Reason are true and Reason is correct explanation of Assertion. b) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion. c) If Assertion is true but Reason is false. d) If both Assertion and Reason are false.		
Q.13	Assertion (A): A negative charge in an electric field moves along the direction of the electric field. Reason (R): On a negative charge a force acts in the direction of the electric field.		1
Q.14.	Assertion (A) : In Young's double slit experiment all fringes are of equal width. Reason (R) : The fringe width depends upon wavelength of light (λ) used, distance of screen from plane of slits (D) and slit separation (d).		1
Q.15.	Assertion (A): The positively charged nucleus of an atom has a radius of almost 10^{-15}m. Reason (R): In a-particle scattering experiment, the distance of closest approach for a-particles is $\approx 10^{-15}\text{m}$.		1
Q.16.	Assertion (A) : Silicon is preferred over Germanium for making semiconductor devices. Reason (R) : The energy gap for Germanium is more than the energy gap for Silicon.		1
SECTION-B (5x2=10 marks)			
Q.17.	Write Maxwell's generalization of Ampere's Circuital law. Show that in the process of charging a capacitor, the current produced within the plates of the capacitor is $i = \epsilon_0 d\phi_E/dt$ where ϕ_E is the electric flux produced during charging of the capacitor plates.		2
Q.18.	Two electric bulbs P and Q have their resistances in the ratio of 1:2. They are connected in series across a battery. Find the ratio of the power dissipation in there bulbs.		2

	OR An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1.5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the both cases be 1A calculate the internal resistance of battery.	
Q.19.	Two charged spherical conductors of radii R_1 and R_2 when connected by conducting wire acquire charges q_1 and q_2 respectively. Find the ratio of their charge densities in terms of their radii	2
Q.20.	A device „X“ is connected to an ac source $V = V_0 \sin \omega t$. The variation of voltage, current and power in one cycle is shown in the following graph (a) Identify the device “X”. (b) Which of the curves A, B and C represent the voltage, current and the power consumed in the circuit? 	2
Q.21.	Draw the circuit diagram for studying the V-I characteristics of a p-n junction diode in (i) forward bias and (ii) reverse bias. Draw the typical V-I characteristics of a silicon diode in both the cases.	2
SECTION-C (7x3=21 marks)		
Q.22.	A point object 'O' is kept in a medium of refractive index n_1 in front of a convex spherical surface of radius of curvature R which separates the second medium of refractive index n_2 from the first one as shown in the figure. Draw the ray diagram showing the image formation and deduce the relationship between the object distance and the image distance in terms of n_1, n_2 and R. 	3
Q.23.	Two large charged plane sheets of charge densities $+\sigma$ and $-\sigma$ C/m² are arranged vertically with a separation of distance 'd' between them. Deduce expressions for the electric field at point (i) to the left of the first sheet, (ii) to the right side of the second sheet and (iii) between the sheets. OR In the figure given below, find the (a) Equivalent capacitance of the network between points A and B. Given : $C_1 = C_5 = 4 \mu\text{F}$, $C_2 = C_3 = C_4 = 2 \mu\text{F}$. (b) maximum charge supplied by the battery, and (c) total energy stored in the network. 	3
Q.24	(a) Sketch the graphs showing variations of stopping potential with frequency of incident radiation for two photosensitive materials A and B having threshold frequencies $\nu_A > \nu_B$.	3

	(b) In which case is the stopping potential more and why? Given that both plate A and B has same incident frequency. (c) Does the slope of the graph depend on the nature of the material used? Explain.	
Q.25.	(a) The figure shows the variation of photoelectric current (I) versus applied voltage (V) for two photosensitive materials and for two different intensities of the incident radiation. Identify the pairs of curves that correspond to different materials for the same intensity of incident radiation. Justify your answer in brief. (b) What is the wavelength of a photon of energy $3.3 \times 10^{-19} \text{ J}$? 	3
Q.26.	The figure shows energy level diagram of hydrogen atom. (i) Find out the transition which results in the emission of a photon of wavelength 496 nm (ii) Which transition corresponds to the emission of radiation of maximum wavelength? Justify your answer. 	3
Q.27.	Draw a labelled circuit diagram of a junction diode as a full wave rectifier. Explain its underlying working. Depict the input and output wave forms.	3
Q.28.	a) Using Biot- Savart law, write the expression for the magnetic field 'B' due to an element 'dl' carrying a current 'I' at a distance 'r' from it in vector form b) A circular coil of 'N' turns and diameter 'd' carries a current 'I'. It is unwound and rewound to make another coil of diameter '2d', current 'I' remaining the same. Calculate the ratio of the magnetic moments of the new coil and the original coil.	3
	SECTION-D (2x4=8 marks) Case Study Based Questions Questions number 29 and 30 are case study based questions. Read the following paragraphs and answer the questions that follow	
Q.29	MOVING COIL GALVANOMETER: The galvanometer is a device used to detect the current flowing in a circuit or a small potential difference applied to it. It consists of a coil with many turns, free to rotate about a fixed axis, in a uniform radial magnetic field formed by using concave pole pieces of a magnet. When current flows through the coil, torque act on it. 	4

	(i) What is the principle of moving coil galvanometer? (a) Torque acting on a current carrying coil placed in a uniform magnetic field. (b) Torque acting on a current carrying coil placed in a non-uniform magnetic field. (c) Potential difference developed in the current carrying coil. (d) None of these.	
	(ii) If the field is radial, then the angle between magnetic moment of galvanometer coil and the magnetic field will be (a) 0° (b) 30° (c) 60° (d) 90°	
	(iii) Why pole pieces are made concave in the moving coil galvanometer? (a) to make the magnetic field radial. (b) to make the magnetic field uniform. (c) to make the magnetic field non-uniform. (d) none of these.	
	(iv) What is the function of radial field in the moving coil galvanometer? (a) to make the torque acting on the coil maximum. (b) to make the magnetic field strong. (c) to make the current scale linear. (d) All the above. OR If the rectangular coil used in the moving coil galvanometer is made circular, then what will be the effect on the maximum torque acting on the coil in magnetic field for the same area of the coil? (a) remains the same (b) becomes less in circular coil (c) becomes greater in circular coil (d) depends on the orientation of the coil	
Q.30	Optical fibres: Now-a-days optical fibres are extensively used for transmitting audio and video signals through long distances. Optical fibres too make use of the phenomenon of total internal reflection. Optical fibres are fabricated with high quality composite glass/quartz fibres. Each fibre consists of a core and cladding. The refractive index of the material of the core is higher than that of the cladding. When a signal in the form of light is directed at one end of the fibre at a suitable angle, it undergoes repeated total internal reflections along the length of the fibre and finally comes out at the other end. Since light undergoes total internal reflection at each stage, there is no appreciable loss in the intensity of the light signal. Optical fibres are fabricated such that light reflected at one side of inner surface strikes the other at an angle larger than the critical angle. Even if the fibre is bent, light can easily travel along its length. Thus, an optical fibre can be used to act as an optical pipe.	4

		
	i) Which of the following statement is not true. a) Optical fibres is based on the principle of total internal reflection. b) The refractive index of the material of the core is less than that of the cladding. c) an optical fibre can be used to act as an optical pipe. d) there is no appreciable loss in the intensity of the light signal while propagating through an optical fibre.	
	ii) What is the condition for total internal reflection to occur? a) angle of incidence must be equal to the critical angle. b) angle of incidence must be less than the critical angle. c) angle of incidence must be greater than the critical angle. d) None of the above.	
	iii) Which of the following is not an application of total internal reflection? a) Mirage b) Sparkling of diamond c) Splitting of white light through a prism. d) Totally reflecting prism.	
	(iv) condition for total internal reflection is . a) light travel from rarer to denser & critical angle smaller than angle of incidence b) light travel from rarer to denser & critical angle grater than angle of incidence c) light travel from denser to rarer & critical angle smaller than angle of incidence d) light travel from denser to rarer & critical angle greater than angle of incidence OR Optical fibres are used extensively to transmit a) Optical Signal b) current c) Sound waves d) None of the above	
	SECTION -E (3x5=15 marks)	
Q.31	(i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone bridge. (ii) Use Kirchhoff's rule to determine the current flowing through the branches MN, TO and SP in the circuit shown in the figure. OR 	5

	(i) Derive the relation between the current density, electric field and the conductivity of a conductor. Briefly explain with graph the variation of conductivity of a conductor with rise in temperature. (ii) Two conducting wires X and Y of same diameter but different materials are joined in series across a battery. If the number density of electrons in X is twice that in Y. find the ratio of drift velocities of electrons in the two wires.	
Q.32.	(i) Draw a labelled diagram of a step-down transformer. Name the principle of its working. Express the turn ratio in terms of voltages. Write one loss of transformer & how it minimises. (ii) How much current is drawn by the primary of a transformer connected to 220 V supply when it delivers power to a 110 V- 550W refrigerator? OR (i) Define coefficient of self-induction. Obtain an expression for self-induction of a long solenoid of length l, area of cross section A having N turns. (ii) The magnetic flux linked with a coil (in Wb) is given by the equation: $\phi = 5t^2 + 3t + 16.$ Find The induced e.m.f. in the coil in the fourth second.	5
Q.33.	(i) Draw a labelled ray diagram showing the image formation by a refracting telescope in normal adjustment condition. Derive its magnifying power. Write two limitations of a refracting telescope over reflecting telescope. (ii) The magnifying power of an refracting telescope is 8 and the distance between the two lenses is 54cm. Find the focal length of eye lens and objective lens. OR (a) Define wavefront. Use Huygens principle to verify the laws of refraction. (b) A parallel beam of light of 500 nm falls on a narrow slit and the resulting diffraction pattern is observed on a screen 1 m away. It is observed that the first minimum is at a distance of 2.5 mm from the centre of the screen. Calculate the width of the slit.	5

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